



Capstone Project for Software Engineering 2

CMU-SE 451

Database Design

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LearnSight: Interactive Real-World English Learning Platform for Children

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PROJECT INFORMATION

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1. Introduction

Database Design is the process of translating a logical data model into a specific physical structure within a chosen Database Management System (DBMS). This process places a strong emphasis on the system's performance requirements, ensuring that abstract data concepts are converted into tangible structures, such as tables, in the database.

1.1 Purpose

The primary purpose of database design is to ensure that every transaction within the system meets or exceeds its established performance benchmarks. This document considers data and transaction volumes to produce an optimal schema and environment that will deliver the necessary performance.

1.2 Scope

Database Design Document has the following objectives:

- To describe the design of the database, which is a collection of related data stored in one or more computer files that can be accessed by users or developers through a DBMS.
- To serve as a foundation for implementing the database and its related software components. It provides the acquirer with visibility into the design and supplies the information necessary for development.

1.3 Introduction about MongoDB

Database Design is the process of translating a logical data model into a specific physical structure within a chosen Database Management System (DBMS). This process places a strong emphasis on the system's performance requirements, ensuring that abstract data concepts are converted into tangible structures, such as tables, in the database.

2. Database Diagram

2.1 Table Overview

Table Name	Short Description
User	This table stores user account information, profiles, roles (user, admin), and credentials.
Story	This table holds the core content of a story, including its title, theme, cover image, and pages.
Page	This table represents a single page within a Story, containing text, an image, and an audio file.
Post	This table shows posts on the Discover feed, used for sharing a Story with the community.
Comment	This table stores user comments and ratings submitted on a Post.
Like	This table records when a User likes a Post, acting as a junction table.
Favorite	This table records when a User adds a Story to their favorites, acting as a junction table.
Message	This table shows private messages sent and received between two users.
Report	This table logs reports submitted by users for inappropriate content.
SystemLog	This table records important system actions for administrative and debugging purposes.

2.2 Entity Relationship Diagram

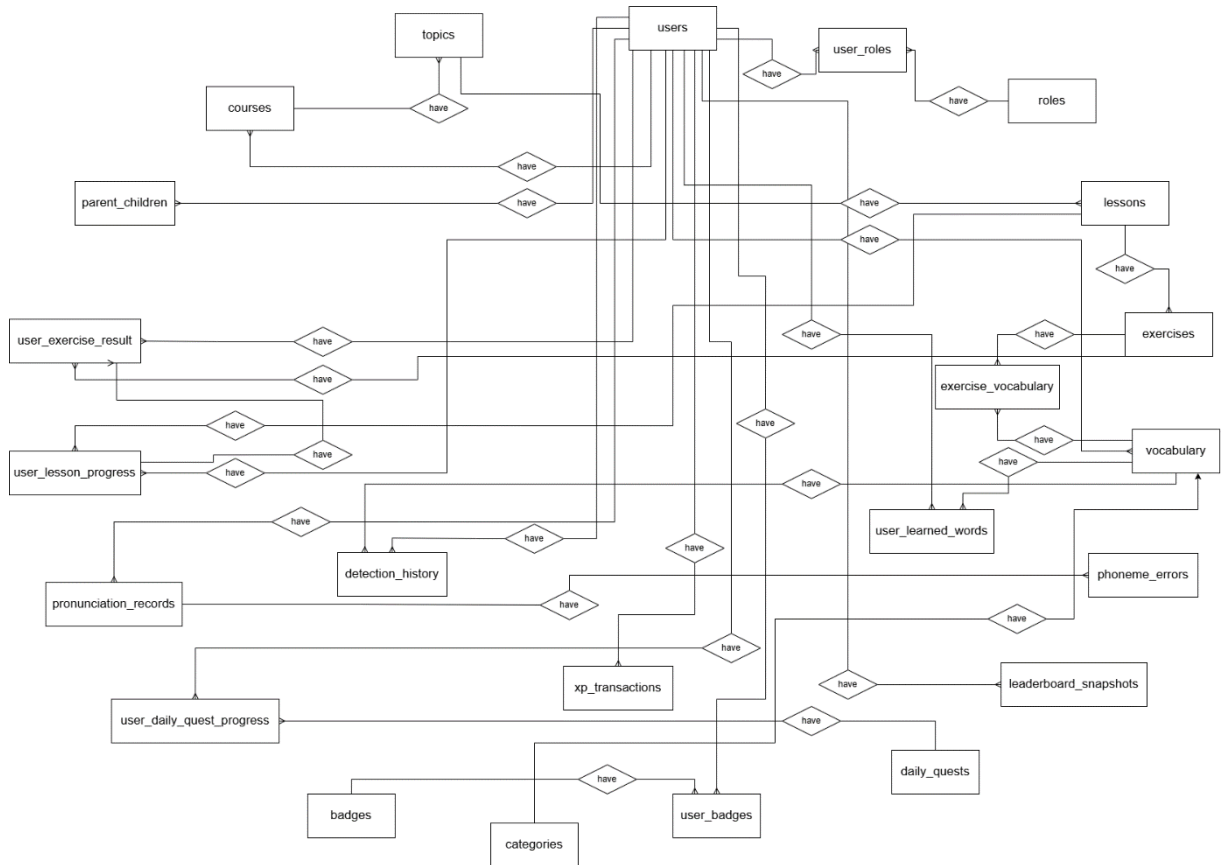


Figure 1. Entity Relationship Diagram

2.3 Table Relationship Diagram

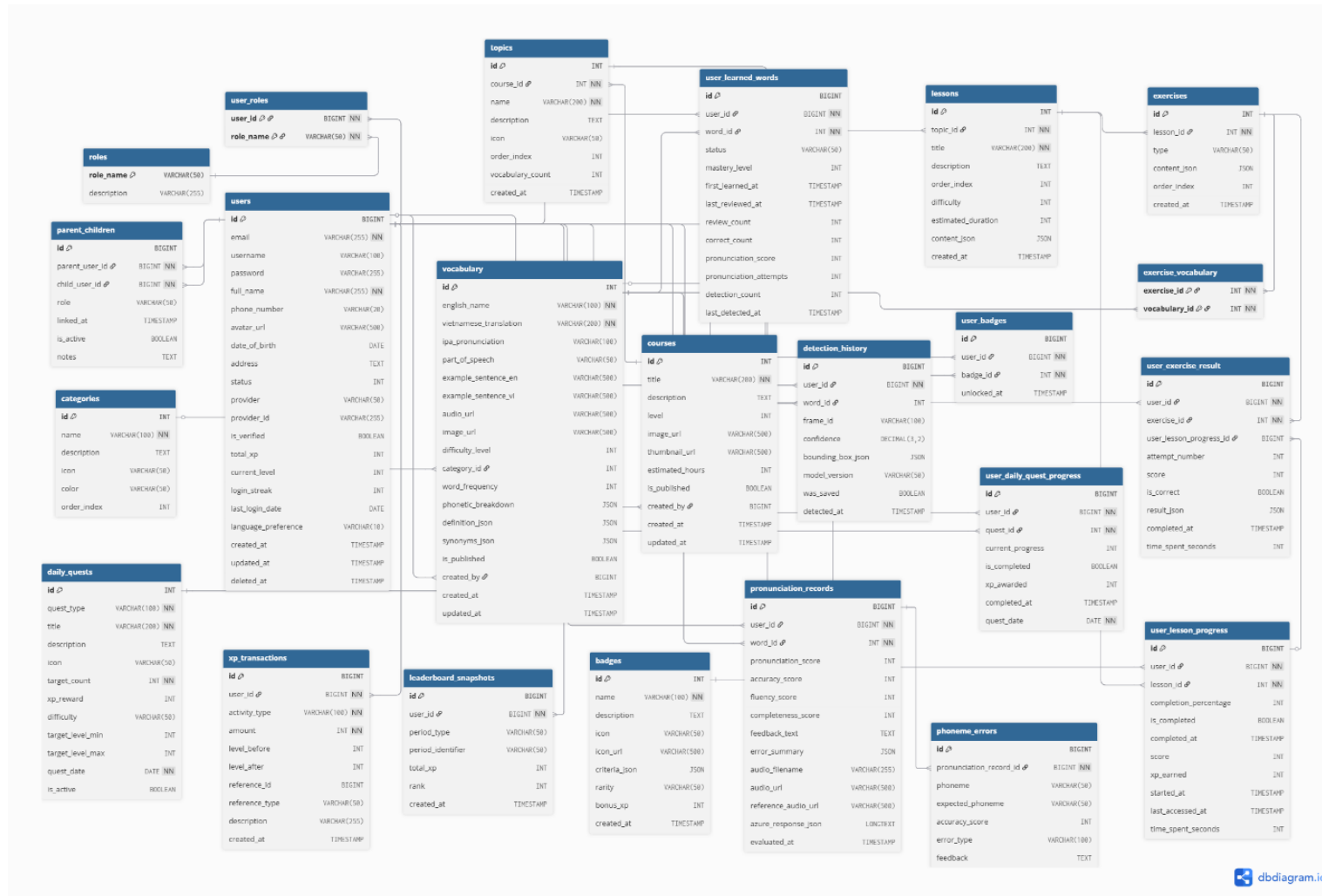


Figure 2. Table Relationship Diagram

3. Database Design for Sprint

3.1. Table user

user				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	The unique identifier for the user.
2	email	VARCHAR(255)	Unique, Not Null	User's unique email, used for login.
3	username	VARCHAR(100)		Optional unique username.
4	password	VARCHAR(255)		Hashed password (bcrypt) for security.
5	full_name	VARCHAR(255)	Not Null	Display full name of the user.
6	phone_number	VARCHAR(20)		User's phone number.
7	avatar_url	VARCHAR(500)		URL for the user's profile picture.
8	date_of_birth	DATE		User's date of birth.
9	address	TEXT		Physical address of the user.
10	status	INT	Default: 1	Account status (0=inactive, 1=active, 2=suspended).
11	provider	VARCHAR(50)	Default: 'LOCAL'	Auth provider (LOCAL, GOOGLE, FACEBOOK).
12	provider_id	VARCHAR(255)		ID from external auth provider.
13	is_verified	BOOLEAN	Default: false	Indicates if email is verified.
14	total_xp	INT	Default: 0	Cumulative experience points earned.
15	current_level	INT	Default: 1	Current gamification level (1-50).
16	login_streak	INT	Default: 0	Consecutive days the user has logged in.
17	last_login_date	DATE		Date of the user's last login.
18	language_preference	VARCHAR(10)	Default: 'vi'	Preferred interface language.
19	created_at	TIMESTAMP		Timestamp when the user was created.
20	updated_at	TIMESTAMP		Timestamp when the user was last updated.

21	deleted_at	TIMESTAMP		Timestamp for soft deletion.
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3.2. Table role

role				
Id	Field	Type	Constraint	Description
1	role_name	VARCHAR(50)	Primary Key	Unique role identifier (e.g., USER, ADMIN).
2	description	VARCHAR(255)		Human-readable role description.

3.3. Table user_roles

user_roles				
Id	Field	Type	Constraint	Description
1	user_id	BIGINT	PK, Foreign Key	Reference to users table.
2	role_name	VARCHAR(50)	PK, Foreign Key	Reference to roles table.

3.4. Table parent_children

parent_children				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique identifier for the relationship.
2	parent_user_id	BIGINT	Foreign Key, Not Null	Reference to the parent's user record.
3	child_user_id	BIGINT	Foreign Key, Not Null	Reference to the child's user record.
4	role	VARCHAR(50)		Role in relationship (PARENT or TEACHER).
5	linked_at	TIMESTAMP		When the connection was established.
6	is_active	BOOLEAN	Default: true	Indicates if the link is currently active.
7	notes	TEXT		Optional notes from parent/teacher.

3.5. Table courses

courses				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Unique identifier for the course.
2	title	VARCHAR(200)	Not Null	Title of the course.
3	description	TEXT		Detailed description of

				course content.
4	level	INT		Course difficulty level (1-10).
5	image_url	VARCHAR(500)		URL for course cover image.
6	thumbnail_url	VARCHAR(500)		URL for smaller thumbnail image.
7	estimated_hours	INT		Estimated time to complete the course.
8	is_published	BOOLEAN	Default: false	Visibility status of the course.
9	created_by	BIGINT	Foreign Key	Admin/Teacher who created this course.
10	created_at	TIMESTAMP		Creation timestamp.
11	updated_at	TIMESTAMP		Last update timestamp.

3.6. Table topics

topics				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Unique identifier for the topic.
2	course_id	INT	Foreign Key, Not Null	Reference to parent course.
3	name	VARCHAR(200)	Not Null	Name of the topic (e.g., Animals).
4	description	TEXT		Topic description.
5	icon	VARCHAR(50)		Emoji or icon class representing the topic.
6	order_index	INT		Display order within the course.
7	vocabulary_count	INT	Default: 0	Cached count of words in this topic.
8	created_at	TIMESTAMP		Creation timestamp.

3.7. Table lessons

lessons				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Unique identifier for the lesson.
2	topic_id	INT	Foreign Key, Not Null	Reference to parent topic.
3	title	VARCHAR(200)	Not Null	Title of the lesson.
4	description	TEXT		Lesson objectives and

				description.
5	order_index	INT		Display order within the topic.
6	difficulty	INT		Difficulty scaling (1-10).
7	estimated_duration	INT		Expected completion time in minutes.
8	content_json	JSON		Flexible JSON for lesson structural data.
9	created_at	TIMESTAMP		Creation timestamp.

3.8. Table exercises

exercises				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Unique identifier for the exercise.
2	lesson_id	INT	Foreign Key, Not Null	Reference to parent lesson.
3	type	VARCHAR(50)		Type (FLASHCARD, LISTENING, SPEAKING, QUIZ).
4	content_json	JSON		Exercise logic and question configuration.
5	order_index	INT		Sequence within the lesson.
6	created_at	TIMESTAMP		Creation timestamp.

3.9. Table categories

categories				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Unique category ID.
2	name	VARCHAR(100)	Not Null	Category name.
3	description	TEXT		Category description.
4	icon	VARCHAR(50)		UI icon for category.
5	color	VARCHAR(50)		UI color theme (hex or tailwind class).
6	order_index	INT		Display sorting order.

3.10. Table vocabulary

vocabulary				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key,	Unique word ID.

			Auto Increment	
2	english_name	VARCHAR(100)	Unique, Not Null	The English word.
3	vietnamese_translation	VARCHAR(200)	Not Null	Vietnamese meaning.
4	ipa_pronunciation	VARCHAR(100)		IPA format (e.g., /'æp.əl/).
5	part_of_speech	VARCHAR(50)	Default: 'NOUN'	Grammar part (NOUN, VERB, etc.).
6	example_sentence_en	VARCHAR(500)		English example usage.
7	example_sentence_vi	VARCHAR(500)		Vietnamese translation of example.
8	audio_url	VARCHAR(500)		S3 URL to native speaker audio.
9	image_url	VARCHAR(500)		S3 URL to illustration image.
10	difficulty_level	INT	Default: 1	Word difficulty (1-10).
11	category_id	INT	Foreign Key	Reference to categories table.
12	word_frequency	INT		Commonality ranking of the word.
13	phonetic_breakdown	JSON		Detailed phoneme array for AI scoring.
14	definition_json	JSON		Multiple dictionary definitions.
15	synonyms_json	JSON		List of related words.
16	is_published	BOOLEAN	Default: true	Visibility flag.
17	created_by	BIGINT	Foreign Key	Admin who added the word.
18	created_at	TIMESTAMP		Creation timestamp.
19	updated_at	TIMESTAMP		Last update timestamp.

3.11. Table exercise_vocabulary

exercise_vocabulary				
Id	Field	Type	Constraint	Description
1	exercise_id	INT	PK, Foreign Key	Reference to exercises table.
2	vocabulary_id	INT	PK, Foreign Key	Reference to vocabulary table.

3.12. Table user_lesson_progress

user_lesson_progress				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key,	Unique record ID.

			Auto Increment	
2	user_id	BIGINT	Foreign Key, Not Null	Reference to users table.
3	lesson_id	INT	Foreign Key, Not Null	Reference to lessons table.
4	completion_percentage	INT	Default: 0	Progress (0-100).
5	is_completed	BOOLEAN	Default: false	True if 100% finished.
6	completed_at	TIMESTAMP		When the lesson was finished.
7	score	INT		Final graded score for the lesson.
8	xp_earned	INT	Default: 0	XP awarded for this specific lesson.
9	started_at	TIMESTAMP		When user first opened the lesson.
10	last_accessed_at	TIMESTAMP		Most recent interaction time.
11	time_spent_seconds	INT	Default: 0	Accumulated active learning time.

3.13. Table user_exercise_result

user_exercise_result				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique record ID.
2	user_id	BIGINT	Foreign Key, Not Null	Reference to users.
3	exercise_id	INT	Foreign Key, Not Null	Reference to exercises.
4	user_lesson_progress_id	BIGINT	Foreign Key	Link to the parent lesson session.
5	attempt_number	INT		Which try this is (1st, 2nd, etc.).
6	score	INT		Score for this exercise component.
7	is_correct	BOOLEAN		Pass/Fail binary status.
8	result_json	JSON		Detailed answers/choices made.
9	completed_at	TIMESTAMP		Completion time.
10	time_spent_seconds	INT		Time taken for this specific exercise.

3.14. Table user_learned_words

user_learned_words				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique record ID.
2	user_id	BIGINT	Foreign Key, Not Null	Reference to users.
3	word_id	INT	Foreign Key, Not Null	Reference to vocabulary.
4	status	VARCHAR(50)	Default: 'NEW'	Enum: NEW, LEARNING, MASTERED.
5	mastery_level	INT	Default: 0	SM-2 algorithm mastery (0-100).
6	first_learned_at	TIMESTAMP		First encounter date.
7	last_reviewed_at	TIMESTAMP		Most recent practice date.
8	review_count	INT	Default: 0	Total times reviewed in flashcards/tests.
9	correct_count	INT	Default: 0	Number of correct recalls.
10	pronunciation_score	INT	Default: 0	Best Azure AI pronunciation score.
11	pronunciation_attempts	INT	Default: 0	Times practiced speaking this word.
12	detection_count	INT	Default: 0	Times found via Camera AI.
13	last_detected_at	TIMESTAMP		Most recent camera detection.

3.15. Table detection_history

detection_history				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique log ID.
2	user_id	BIGINT	Foreign Key, Not Null	The user who scanned.
3	word_id	INT	Foreign Key	Recognized word (Null if unknown).
4	frame_id	VARCHAR(100)		Identifier for the video frame.
5	confidence	DECIMAL(3,2)		YOLO AI confidence score (0.00-1.00).
6	bounding_box_json	JSON		Coordinates of object in

				frame.
7	model_version	VARCHAR(50)		Tracking AI model used (e.g., YOLOv11n).
8	was_saved	BOOLEAN	Default: false	True if user manually saved the word.
9	detected_at	TIMESTAMP		Scan timestamp.

3.16. Table pronunciation_records

pronunciation_records				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique evaluation record.
2	user_id	BIGINT	Foreign Key, Not Null	Reference to user.
3	word_id	INT	Foreign Key, Not Null	Reference to evaluated word.
4	pronunciation_score	INT		Overall Azure score (0-100).
5	accuracy_score	INT		Azure accuracy metric (0-100).
6	fluency_score	INT		Azure fluency metric (0-100).
7	completeness_score	INT		Azure completeness metric (0-100).
8	feedback_text	TEXT		AI-generated improvement advice.
9	error_summary	JSON		Parsed error types for frontend rendering.
10	audio_filename	VARCHAR(255)		S3 storage key for user's recording.
11	audio_url	VARCHAR(500)		Public link to playback recording.
12	reference_audio_url	VARCHAR(500)		Link to native audio compared against.
13	azure_response_json	LONGTEXT		Raw response from Azure API for debug.
14	evaluated_at	TIMESTAMP		Time of processing.

3.17. Table phoneme_errors

phoneme_errors				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique error ID.

2	pronunciation_record_id	BIGINT	Foreign Key, Not Null	Link to parent evaluation.
3	phoneme	VARCHAR(50)		The specific sound user failed on.
4	expected_phoneme	VARCHAR(50)		The correct IPA sound.
5	accuracy_score	INT		Score for this specific phoneme.
6	error_type	VARCHAR(100)		Classification (e.g., Mispronunciation).
7	feedback	TEXT		Specific tip for this sound.

3.18. Table xp_transactions

xp_transactions				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Ledger entry ID.
2	user_id	BIGINT	Foreign Key, Not Null	User receiving XP.
3	activity_type	VARCHAR(100)	Not Null	Action (LEARN_WORD, DAILY_QUEST, etc.).
4	amount	INT	Not Null	XP gained (or lost).
5	level_before	INT		User level prior to transaction.
6	level_after	INT		User level after transaction.
7	reference_id	BIGINT		ID of related entity (lesson, quest).
8	reference_type	VARCHAR(50)		Table name of the reference_id.
9	description	VARCHAR(255)		Human-readable log message.
10	created_at	TIMESTAMP		Transaction time.

3.19. Table daily_quests

daily_quests				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Quest instance ID.
2	quest_type	VARCHAR(100)	Not Null	Classification (e.g., CAMERA_DETECTION).
3	title	VARCHAR(200)	Not Null	Display title.
4	description	TEXT		Instructions for user.
5	icon	VARCHAR(50)		Visual representation.

6	target_count	INT	Not Null	Goal number to reach.
7	xp_reward	INT	Default: 50	Bounty for completion.
8	difficulty	VARCHAR(50)	Default: 'MEDIUM'	EASY, MEDIUM, HARD.
9	target_level_min	INT	Default: 1	Minimum user level to receive quest.
10	target_level_max	INT	Default: 50	Maximum user level to receive quest.
11	quest_date	DATE	Not Null	The active day for this quest.
12	is_active	BOOLEAN	Default: true	Can be disabled by admins.

3.20. Table user_daily_quest_progress

user_daily_quest_progress				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Unique tracking ID.
2	user_id	BIGINT	Foreign Key, Not Null	Reference to user.
3	quest_id	INT	Foreign Key, Not Null	Reference to daily_quests.
4	current_progress	INT	Default: 0	Current count towards target.
5	is_completed	BOOLEAN	Default: false	True if target_count is met.
6	xp_awarded	INT	Default: 0	Actual XP given (may include streak bonus).
7	completed_at	TIMESTAMP		Completion timestamp.
8	quest_date	DATE	Not Null	Historical tracking date.

3.21. Table badges

badges				
Id	Field	Type	Constraint	Description
1	id	INT	Primary Key, Auto Increment	Unique badge definition.
2	name	VARCHAR(100)	Not Null	Display name.
3	description	TEXT		How to earn it.
4	icon	VARCHAR(50)		Emoji fallback.
5	icon_url	VARCHAR(500)		S3 URL for high-res badge art.
6	criteria_json	JSON		Logic rules for backend validation.
7	rarity	VARCHAR(50)		COMMON, RARE, EPIC, LEGENDARY.
8	bonus_xp	INT		Extra reward for

				unlocking.
9	created_at	TIMESTAMP		Definition creation date.

3.22. Table user_badges

user_badges				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Award record ID.
2	user_id	BIGINT	Foreign Key, Not Null	Awarded user.
3	badge_id	INT	Foreign Key, Not Null	The badge earned.
4	unlocked_at	TIMESTAMP		When the milestone was reached.

3.23. Table leaderboard_snapshots

leaderboard_snapshots				
Id	Field	Type	Constraint	Description
1	id	BIGINT	Primary Key, Auto Increment	Snapshot record ID.
2	user_id	BIGINT	Foreign Key, Not Null	Reference to user.
3	period_type	VARCHAR(50)	Not Null	WEEK, MONTH, or ALLTIME.
4	period_identifier	VARCHAR(50)	Not Null	e.g., '2026-W10' or '2026-03'.
5	total_xp	INT	Default: 0	Score locked in for this period.
6	rank	INT		Final leaderboard position.
7	created_at	TIMESTAMP	Default: now()	When the snapshot was calculated.

4. Hardware and software Requirements

This section provides an overview of hardware and software requirements. Below are descriptions of the technological components of the Smart Comic Drawing App for Kids:

Attributes of Easy English WEBSITE	
Attributes	Descriptions
Database	MySQL
Software	Reactjs, Spring Boot, Redis, Websocket
Hardware	Computer

Library	Hook, Redux, Fontawesome, Material ui, Ant Design
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